



Kauno technologijos universitetas

Informatikos fakultetas

T120B165 Saityno taikomųjų programų projektavimas

Projekto „Vaistų valdymo sistema“ ataskaita

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Dėstytojas

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1. Sprendžiamo uždavinio aprašymas

1.1. Sistemos paskirtis

„Vaistų valdymo sistema“ yra skirta palengvinti turimų vaistų sekimą, valdymą bei vartojimą.

Ši sistema veikia internetinėje svetainėje. Naudotojas, norėdamas naudotis šia sistema, prisiregistruoja prie svetainės, susikuria naują vaistų rezervą ir iš duomenų bazės pasirenka kokius vaistus nori pridėti, nurodydamas jų kiekį ir galiojimo datą. Pridėjęs vaistus, naudotojas gali jų kiekį redaguoti bei pradėti vaistų vartojimo laikotarpį, nurodant vartojimo dažnį bei terminą. Taip pat, naudotojas, neradęs reikiamų vaistų duomenų bazėje, juos gali pridėti pats sau lokaliai arba globaliai, kad būtų prieinami visiems naudotojams.

Globaliai pridedamus vaistus turi patvirtinti administratorius, kuris taip pat gali juos koreguoti ir šalinti.

1.2. Funkciniai reikalavimai

Neregistruotas sistemos naudotojas gali:

1. Registruotis
2. Peržiūrėti duomenų bazėje esančius vaistus

Registruotas sistemos naudotojas gali:

1. Prisijungti
2. Atsijungti
3. Sukurti naują vaistų rezervą
4. Peržiūrėti vaistų rezervą
5. Pridėti vaistus prie vaistų rezervo
6. Pridėti naujus vaistus į duomenų bazę
 - a. Lokaliai, pasiekama tik naudotojui
 - b. Globaliai, pasiekama visiems naudotojams
7. Redaguoti vaistus pridėtus į vaistų rezervą
8. Šalinti vaistus iš vaistų rezervo
9. Pradėti vaistų vartojimo laikotarpį
10. Užbaigti vaistų vartojimo laikotarpį
11. Redaguoti vaistų vartojimo laikotarpį
12. Peržiūrėti vaistų vartojimo laikotarpį
13. Peržiūrėti vaistų kategorijų sąrašą
14. Peržiūrėti vaistų sąrašą

Administratorius gali:

1. Pridėti naujus vaistus į duomenų bazę globaliai
2. Redaguoti vaistus esančius duomenų bazėje
3. Šalinti vaistus esančius duomenų bazėje
4. Peržiūrėti duomenų bazėje esančių vaistų informaciją
5. Atsakyti į užklausas pridėti vaistus į globalią duomenų bazę
6. Pridėti naują vaistų kategoriją
7. Peržiūrėti visas vaistų kategorijas
8. Redaguoti vaistų kategoriją
9. Šalinti vaistų kategoriją

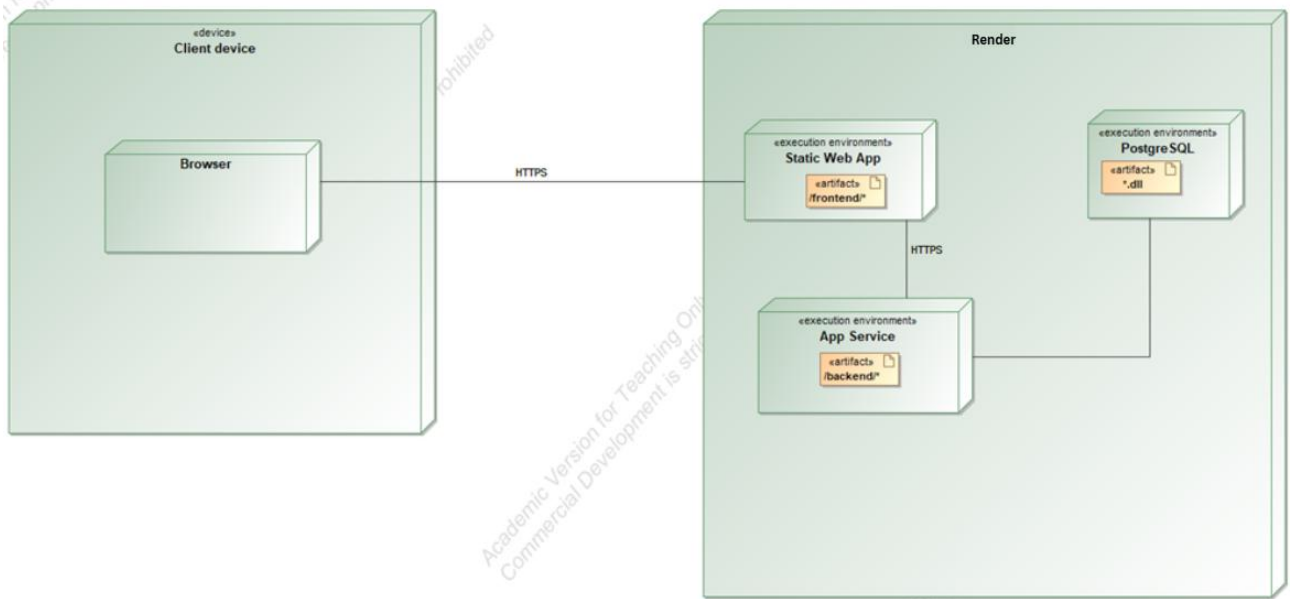
10. Peržiūrėti vaistų sąrašą

2. Pasirinktų technologijų aprašymas

Projektas susidės iš 3 pagrindinių dalių:

1. Klientinė dalis (Frontend). Klientinė dalis bus realizuojama pasinaudojant React.js.
2. Serverio dalis (Backend). Serverio dalis bus realizuojama pasinaudojant Django.
3. Duomenų bazė. Bus naudojama PostgreSQL duomenų bazė.

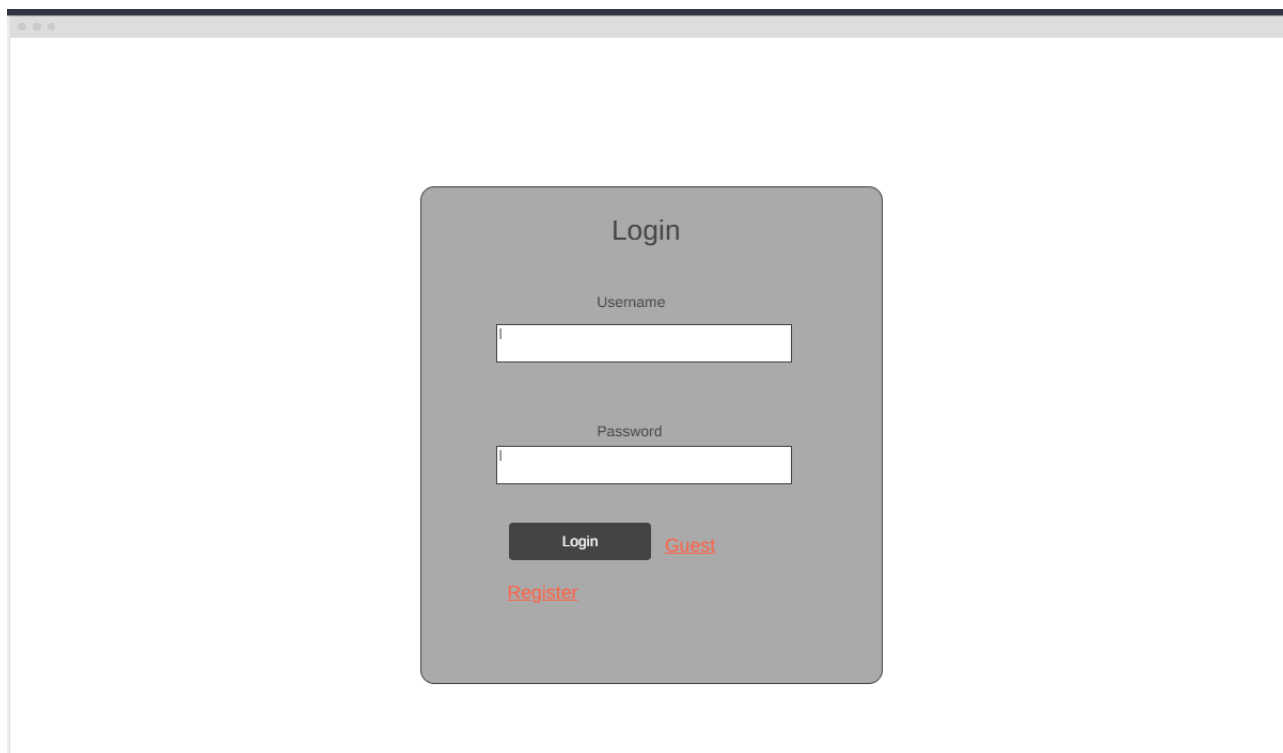
Diegimo diagrama:



1 pav. Diegimo diagrama

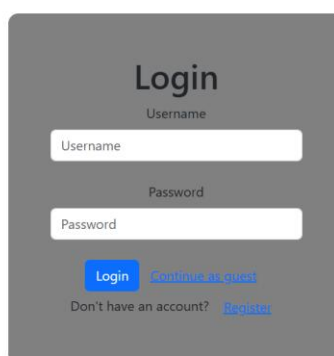
3. Naudotojo sąsajos projektas

3.1. Prisijungimo langas



A wireframe of a login form within a browser window. The form is a gray rounded rectangle with the title "Login" at the top. Below the title are two input fields: "Username" and "Password". Under the "Password" field is a dark gray "Login" button. To the right of the button is a red "Guest" link. Below the button is a red "Register" link.

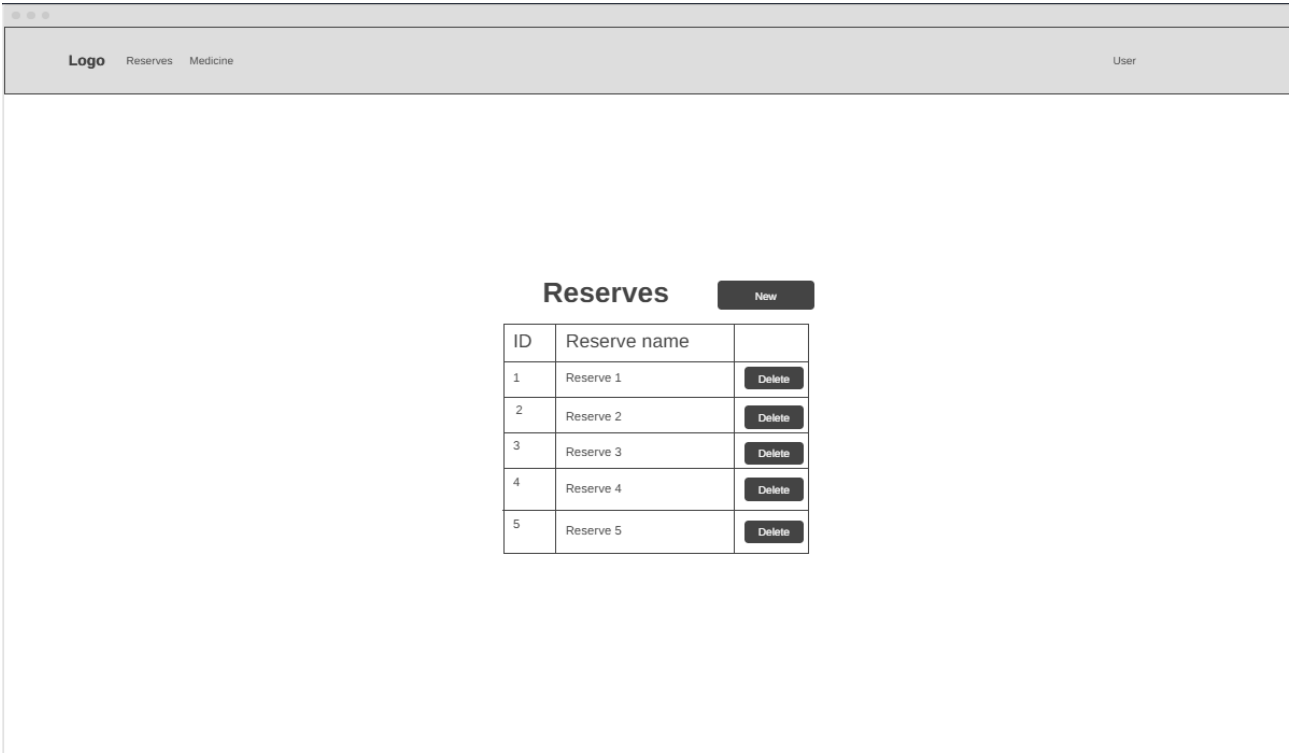
2 pav. Prisijungimo lango wireframe



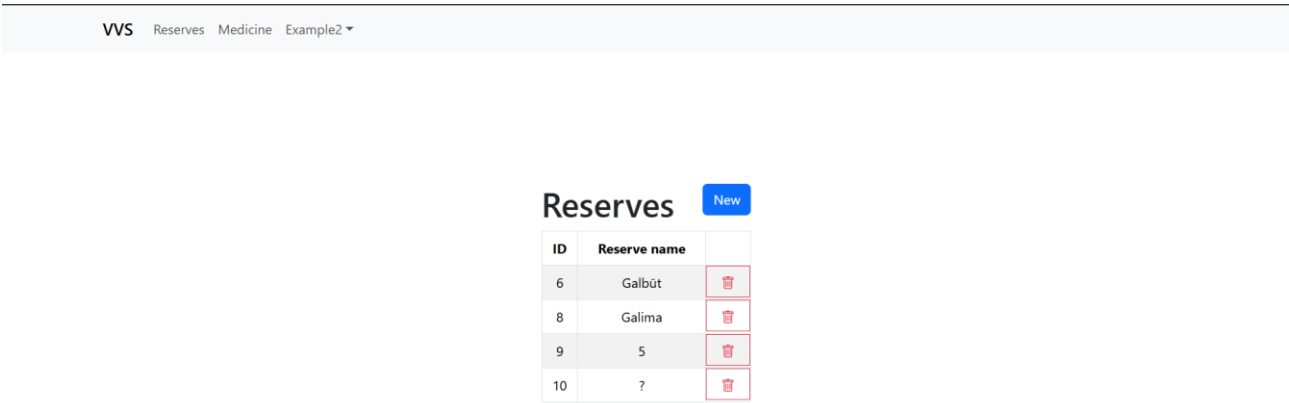
The final design of the login form. It has a dark gray background. The title "Login" is in white. Below it, the "Username" label is in white, followed by a white input field. The "Password" label is in white, followed by a white input field. Below the password field is a blue "Login" button and a blue "Continue as guest" link. At the bottom, the text "Don't have an account?" is in white, followed by a blue "Register" link.

3 pav. Prisijungimo langas

3.2. Rezervų sąrašo langas

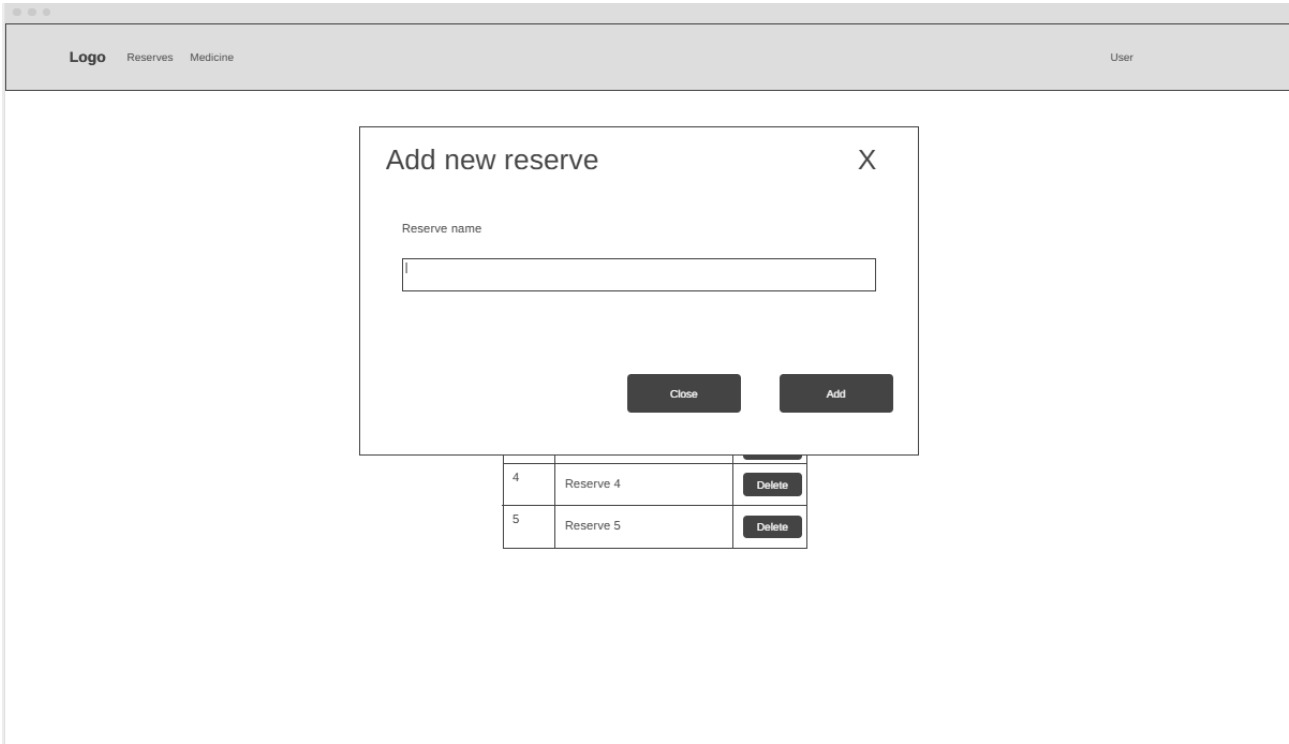


4 pav. Rezervų sąrašo wireframe

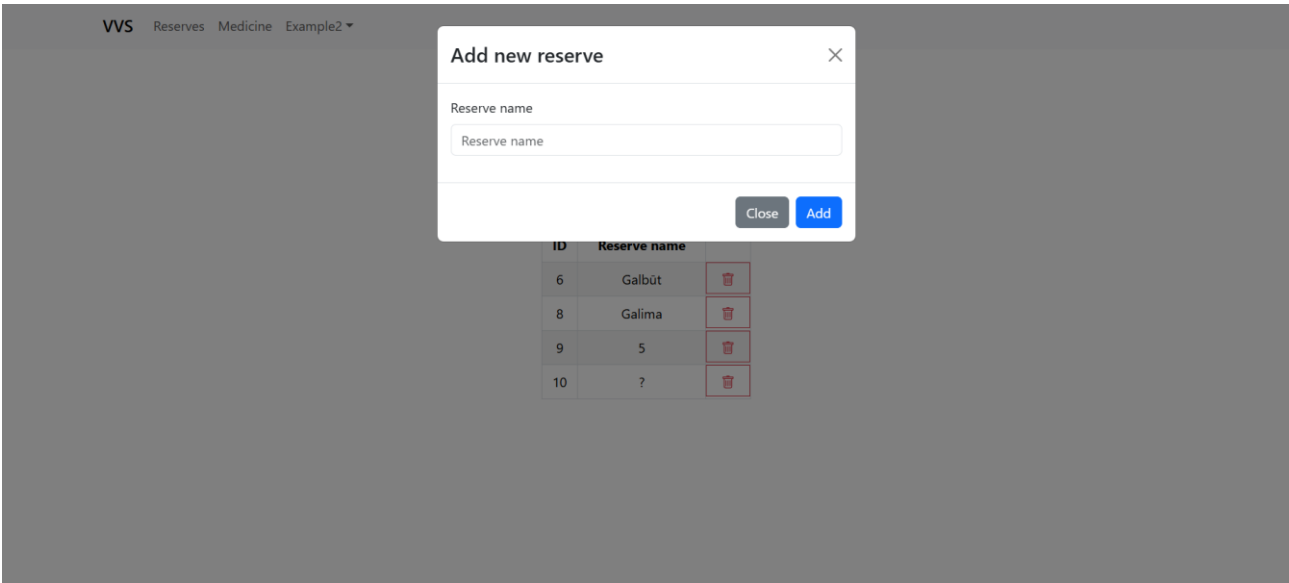


5 pav. Rezervų sąrašas

3.3. Rezervo pridėjimo modalas



6 pav. Rezervo pridėjimo modalo wireframe



7 pav. Rezervo pridėjimo modalas

4. API specifikacija

```
openapi: 3.0.3
info:
  title: ''
  version: 0.0.0
paths:
  /api/category:
    get:
      operationId: api_category_list
      description: Retrieve a list of all available medicine categories.
      summary: List all categories
      tags:
        - api
      responses:
        '200':
          content:
            application/json:
              schema:
                type: array
                items:
                  $ref: '#/components/schemas/Categories'
          description: ''
  /api/category/{id}:
    get:
      operationId: api_category_retrieve
      description: Retrieve details of a specific category. Only accessible to administrators.
      summary: Retrieve category details
      parameters:
        - in: path
          name: id
          schema:
            type: integer
          required: true
      tags:
        - api
      responses:
        '200':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Category'
          description: ''
        '401':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
          description: Not authenticated
        '403':
          content:
            application/json:
              schema:
```

```

        $ref: '#/components/schemas/Error'
      description: Forbidden
    '404':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Not found
  /api/category/{id}/medicine:
    get:
      operationId: api_category_medicine_list
      description: Retrieve all medicines belonging to a specific category.
      summary: List medicines in a category
      parameters:
        - in: path
          name: id
          schema:
            type: integer
          required: true
      tags:
        - api
      responses:
        '200':
          content:
            application/json:
              schema:
                type: array
                items:
                  $ref: '#/components/schemas/Medicine'
          description: ''
        '404':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
          description: Not found
  /api/category/create:
    post:
      operationId: api_category_create_create
      description: Create a new medicine category. Administrator-only endpoint.
      summary: Create a new category
      tags:
        - api
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Category'
          application/x-www-form-urlencoded:
            schema:
              $ref: '#/components/schemas/Category'
          multipart/form-data:
            schema:
              $ref: '#/components/schemas/Category'

```

```

    required: true
  responses:
    '201':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Category'
      description: ''
    '400':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Bad request
    '401':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Not authenticated
    '403':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Forbidden
  /api/category/delete/{id}:
    delete:
      operationId: api_category_delete_destroy
      description: Delete a specific category. Administrator-only endpoint.
      summary: Delete a category
      parameters:
        - in: path
          name: id
          schema:
            type: integer
          required: true
      tags:
        - api
      responses:
        '204':
          description: Category deleted successfully.
        '401':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
          description: Not authenticated
        '403':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
          description: Forbidden

```

```

    '404':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Not found
/api/category/update/{id}:
  put:
    operationId: api_category_update_update
    description: Update an existing category. Administrator-only endpoint.
    summary: Update a category
    parameters:
      - in: path
        name: id
        schema:
          type: integer
        required: true
    tags:
      - api
    requestBody:
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Category'
        application/x-www-form-urlencoded:
          schema:
            $ref: '#/components/schemas/Category'
        multipart/form-data:
          schema:
            $ref: '#/components/schemas/Category'
      required: true
    responses:
      '200':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Category'
        description: ''
      '400':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
        description: Bad request
      '401':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
        description: Not authenticated
      '403':
        content:
          application/json:
            schema:

```

```

        $ref: '#/components/schemas/Error'
      description: Forbidden
    '404':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
          description: Not found
  /api/dosage/{id}:
    get:
      operationId: api_dosage_retrieve
      summary: Retrieve details of a dosage
      parameters:
        - in: path
          name: id
          schema:
            type: integer
          required: true
      tags:
        - api
      responses:
        '200':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Dosage'
              description: ''
        '404':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
              description: Not found
  /api/dosage/create:
    post:
      operationId: api_dosage_create_create
      summary: Create a new dosage entry
      tags:
        - api
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Dosage'
          application/x-www-form-urlencoded:
            schema:
              $ref: '#/components/schemas/Dosage'
          multipart/form-data:
            schema:
              $ref: '#/components/schemas/Dosage'
        required: true
      responses:
        '201':
          content:

```

```

        application/json:
          schema:
            $ref: '#/components/schemas/Dosage'
        description: ''
      '400':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
        description: Bad request
      '401':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
        description: Not authenticated
    /api/dosage/delete/{id}:
      delete:
        operationId: api_dosage_delete_destroy
        summary: Delete an existing dosage
        parameters:
          - in: path
            name: id
            schema:
              type: integer
            required: true
        tags:
          - api
        responses:
          '204':
            description: Dosage deleted successfully
          '401':
            content:
              application/json:
                schema:
                  $ref: '#/components/schemas/Error'
            description: Not authenticated
          '403':
            content:
              application/json:
                schema:
                  $ref: '#/components/schemas/Error'
            description: Forbidden
          '404':
            content:
              application/json:
                schema:
                  $ref: '#/components/schemas/Error'
            description: Not found
    /api/dosage/update/{id}:
      put:
        operationId: api_dosage_update_update
        summary: Update an existing dosage
        parameters:

```

```

- in: path
  name: id
  schema:
    type: integer
  required: true
tags:
- api
requestBody:
  content:
    application/json:
      schema:
        $ref: '#/components/schemas/Dosage'
    application/x-www-form-urlencoded:
      schema:
        $ref: '#/components/schemas/Dosage'
    multipart/form-data:
      schema:
        $ref: '#/components/schemas/Dosage'
  required: true
responses:
  '200':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Dosage'
    description: ''
  '400':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Bad request
  '401':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Not authenticated
  '403':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Forbidden
  '404':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Not found
/api/login:
  post:
    operationId: api_login_create
    description: Authenticate a user with username and password.

```



```

summary: Log in a user
tags:
- api
security:
- {}
responses:
  '200':
    description: User logged in successfully
  '400':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
        description: Bad request
/api/logout:
  post:
    operationId: api_logout_create
    description: Log out the currently authenticated user.
    summary: Log out the current user
    tags:
    - api
    responses:
      '200':
        description: User logged out successfully
      '400':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
            description: Bad request
/api/medicine:
  get:
    operationId: api_medicine_list
    summary: List all medicines
    tags:
    - api
    responses:
      '200':
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: '#/components/schemas/Medicine'
            description: ''
/api/medicine/{id}:
  get:
    operationId: api_medicine_retrieve
    summary: Retrieve a specific medicine
    parameters:
    - in: path
      name: id
      schema:
        type: integer

```

```

      required: true
tags:
- api
responses:
  '200':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Medicine'
    description: ''
  '404':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Not found
/api/medicine/{id}/usage:
get:
  operationId: api_medicine_usage_list
  summary: List usages for a specific medicine
  parameters:
    - in: path
      name: id
      schema:
        type: integer
      required: true
tags:
- api
responses:
  '200':
    content:
      application/json:
        schema:
          type: array
          items:
            $ref: '#/components/schemas/Usage'
    description: ''
  '404':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Not found
/api/medicine/create:
post:
  operationId: api_medicine_create_create
  summary: Create a new medicine
tags:
- api
requestBody:
  content:
    application/json:
      schema:
        $ref: '#/components/schemas/Medicine'

```

```

    application/x-www-form-urlencoded:
      schema:
        $ref: '#/components/schemas/Medicine'
    multipart/form-data:
      schema:
        $ref: '#/components/schemas/Medicine'
  required: true
responses:
  '201':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Medicine'
    description: ''
  '400':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Bad request
  '401':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Not authenticated
/api/medicine/delete/{id}:
  delete:
    operationId: api_medicine_delete_destroy
    summary: Delete a medicine
    parameters:
      - in: path
        name: id
        schema:
          type: integer
        required: true
    tags:
      - api
    responses:
      '204':
        description: Deleted successfully
      '401':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
        description: Not authenticated
      '403':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
        description: Forbidden
      '404':

```

```

        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
        description: Not found
/api/medicine/update/{id}:
  put:
    operationId: api_medicine_update_update
    summary: Update an existing medicine
    parameters:
      - in: path
        name: id
        schema:
          type: integer
        required: true
    tags:
      - api
    requestBody:
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Medicine'
        application/x-www-form-urlencoded:
          schema:
            $ref: '#/components/schemas/Medicine'
        multipart/form-data:
          schema:
            $ref: '#/components/schemas/Medicine'
      required: true
    responses:
      '200':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Medicine'
        description: ''
      '400':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
        description: Bad request
      '401':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
        description: Not authenticated
      '403':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
        description: Forbidden

```

```

    '404':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
          description: Not found
/api/medicine_intake/{id}:
  get:
    operationId: api_medicine_intake_retrieve
    summary: Retrieve details of a medicine intake
    parameters:
      - in: path
        name: id
        schema:
          type: integer
        required: true
    tags:
      - api
    responses:
      '200':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/MedicineIntake'
            description: ''
      '401':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
            description: Not authenticated
      '403':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
            description: Forbidden
      '404':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
            description: Not found
/api/medicine_intake/create:
  post:
    operationId: api_medicine_intake_create_create
    summary: Create a new medicine intake
    tags:
      - api
    requestBody:
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/MedicineIntake'

```

```

    application/x-www-form-urlencoded:
      schema:
        $ref: '#/components/schemas/MedicineIntake'
    multipart/form-data:
      schema:
        $ref: '#/components/schemas/MedicineIntake'
  required: true
responses:
  '201':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/MedicineIntake'
    description: ''
  '400':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Bad request
  '401':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Not authenticated
/api/medicine_intake/delete/{id}:
  delete:
    operationId: api_medicine_intake_delete_destroy
    summary: Delete a medicine intake
    parameters:
      - in: path
        name: id
        schema:
          type: integer
        required: true
    tags:
      - api
    responses:
      '204':
        description: Medicine intake deleted successfully
      '401':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
        description: Not authenticated
      '403':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
        description: Forbidden
      '404':

```

```

        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
        description: Not found
/api/medicine_intake/notifications:
  get:
    operationId: api_medicine_intake_notifications_list
    description: Checks what time of the day it is and retrieves medicine intakes
      which have a dosage set for the current time of the day and medicine has not
      been consumed today on this time period.
    summary: Retrieve medicine that should be taken right now
    tags:
      - api
    responses:
      '200':
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: '#/components/schemas/MedicineIntake'
            description: ''
      '401':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
            description: Not authenticated
/api/medicine_intake/update/{id}:
  put:
    operationId: api_medicine_intake_update_update
    summary: Update a medicine intake
    parameters:
      - in: path
        name: id
        schema:
          type: integer
        required: true
    tags:
      - api
    requestBody:
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/MedicineIntake'
        application/x-www-form-urlencoded:
          schema:
            $ref: '#/components/schemas/MedicineIntake'
        multipart/form-data:
          schema:
            $ref: '#/components/schemas/MedicineIntake'
        required: true
    responses:

```

```

'200':
  content:
    application/json:
      schema:
        $ref: '#/components/schemas/MedicineIntake'
      description: ''
'400':
  content:
    application/json:
      schema:
        $ref: '#/components/schemas/Error'
      description: Bad request
'401':
  content:
    application/json:
      schema:
        $ref: '#/components/schemas/Error'
      description: Not authenticated
'403':
  content:
    application/json:
      schema:
        $ref: '#/components/schemas/Error'
      description: Forbidden
'404':
  content:
    application/json:
      schema:
        $ref: '#/components/schemas/Error'
      description: Not found
/api/medicine_intakes:
  get:
    operationId: api_medicine_intakes_list
    summary: List all medicine intakes for the authenticated user
    tags:
      - api
    responses:
      '200':
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: '#/components/schemas/MedicineIntake'
            description: ''
      '401':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
            description: Not authenticated
/api/pending/medicine:
  get:
    operationId: api_pending_medicine_list

```



```

summary: List all pending medicines
tags:
- api
responses:
  '200':
    content:
      application/json:
        schema:
          type: array
          items:
            $ref: '#/components/schemas/PendingMedicine'
    description: ''
/api/pending/medicine/{id}:
get:
  operationId: api_pending_medicine_retrieve
  summary: Retrieve details of a pending medicine
  parameters:
    - in: path
      name: id
      schema:
        type: integer
      required: true
  tags:
    - api
  responses:
    '200':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/PendingMedicine'
      description: ''
    '404':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Not found
/api/pending/medicine/create:
post:
  operationId: api_pending_medicine_create_create
  summary: Submit a pending medicine for approval
  tags:
    - api
  requestBody:
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/PendingMedicine'
      application/x-www-form-urlencoded:
        schema:
          $ref: '#/components/schemas/PendingMedicine'
      multipart/form-data:
        schema:
          $ref: '#/components/schemas/PendingMedicine'

```

```

    required: true
  responses:
    '201':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/PendingMedicine'
      description: ''
    '400':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Bad request
    '401':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Not authenticated
  /api/pending/medicine/update/{id}:
    put:
      operationId: api_pending_medicine_update_update
      summary: Update a pending medicine
      parameters:
        - in: path
          name: id
          schema:
            type: integer
          required: true
      tags:
        - api
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/PendingMedicine'
          application/x-www-form-urlencoded:
            schema:
              $ref: '#/components/schemas/PendingMedicine'
          multipart/form-data:
            schema:
              $ref: '#/components/schemas/PendingMedicine'
        required: true
      responses:
        '200':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/PendingMedicine'
          description: ''
        '400':
          content:
            application/json:

```

```

        schema:
          $ref: '#/components/schemas/Error'
      description: Bad request
    '401':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Not authenticated
    '403':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Forbidden
    '404':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Not found
  /api/profile:
    get:
      operationId: api_profile_retrieve
      description: Retrieve the profile of the currently authenticated user.
      summary: Get current user profile
      tags:
        - api
      responses:
        '200':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Profile'
          description: ''
        '401':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
          description: Not authenticated
  /api/profile/change_password:
    put:
      operationId: api_profile_change_password_update
      description: Change the password of the currently authenticated user.
      summary: Change user password
      tags:
        - api
      responses:
        '200':
          description: Password changed successfully
        '400':
          content:
            application/json:

```

```

        schema:
          $ref: '#/components/schemas/Error'
      description: Bad request
    '401':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Not authenticated
  /api/profile/update:
    put:
      operationId: api_profile_update_update
      description: Update the profile of the currently authenticated user.
      summary: Update user profile
      tags:
        - api
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Profile'
          application/x-www-form-urlencoded:
            schema:
              $ref: '#/components/schemas/Profile'
          multipart/form-data:
            schema:
              $ref: '#/components/schemas/Profile'
      responses:
        '200':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Profile'
          description: ''
        '400':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
          description: Bad request
        '401':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
          description: Not authenticated
  /api/refresh/:
    post:
      operationId: api_refresh_create
      description: Refrest access token
      summary: Refresh token
      tags:
        - api
      security:

```

```

- {}
responses:
  '200':
    description: Token refreshed successfully
  '400':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Bad request
/api/register:
  post:
    operationId: api_register_create
    description: Create a new user account along with a profile.
    summary: Register a new user
    tags:
      - api
    requestBody:
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/User'
        application/x-www-form-urlencoded:
          schema:
            $ref: '#/components/schemas/User'
        multipart/form-data:
          schema:
            $ref: '#/components/schemas/User'
      required: true
    security:
      - {}
    responses:
      '201':
        description: User created successfully
      '400':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
        description: Bad request
/api/reserve/{id}:
  get:
    operationId: api_reserve_retrieve
    summary: Get details of a reserve
    parameters:
      - in: path
        name: id
        schema:
          type: integer
        required: true
    tags:
      - api
    responses:
      '200':

```

```

    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Reserve'
    description: ''
  '400':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Bad request
  '401':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Not authenticated
  '403':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Forbidden
  '404':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Not found
/api/reserve/{id}/activity:
  get:
    operationId: api_reserve_activity_list
    summary: List reserve consumption activities
    parameters:
      - in: path
        name: id
        schema:
          type: integer
        required: true
    tags:
      - api
    responses:
      '200':
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: '#/components/schemas/ReserveActivity'
        description: ''
      '400':
        content:
          application/json:
            schema:

```

```

        $ref: '#/components/schemas/Error'
      description: Bad request
    '401':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Not authenticated
    '403':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Forbidden
    '404':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Not found
  /api/reserve/{id}/medicine:
    get:
      operationId: api_reserve_medicine_list
      summary: List all medicines in a reserve
      parameters:
        - in: path
          name: id
          schema:
            type: integer
          required: true
      tags:
        - api
      responses:
        '200':
          content:
            application/json:
              schema:
                type: array
                items:
                  $ref: '#/components/schemas/ReserveMedicine'
          description: ''
        '400':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
          description: Bad request
        '401':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
          description: Not authenticated
        '403':

```

```

    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Forbidden
  '404':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Not found
/api/reserve/{id}/medicine/{med_pk}:
get:
  operationId: api_reserve_medicine_retrieve
  summary: Get details of a medicine in a reserve
  parameters:
    - in: path
      name: id
      schema:
        type: integer
      required: true
    - in: path
      name: med_pk
      schema:
        type: integer
      required: true
  tags:
    - api
  responses:
    '200':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/ReserveMedicine'
      description: ''
    '400':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Bad request
    '401':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Not authenticated
    '403':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Forbidden
    '404':

```



```

        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
        description: Not found
/api/reserve/{id}/medicine/{med_pk}/consume:
  put:
    operationId: api_reserve_medicine_consume_update
    description: Consumes a specified amount of medicine and logs the activity.
    summary: Consume medicine from a reserve
    parameters:
      - in: path
        name: id
        schema:
          type: integer
        required: true
      - in: path
        name: med_pk
        schema:
          type: integer
        required: true
    tags:
      - api
    requestBody:
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/ReserveMedicine'
        application/x-www-form-urlencoded:
          schema:
            $ref: '#/components/schemas/ReserveMedicine'
        multipart/form-data:
          schema:
            $ref: '#/components/schemas/ReserveMedicine'
      required: true
    responses:
      '200':
        description: Medicine consumed successfully
      '400':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
        description: Bad request
      '401':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
        description: Not authenticated
      '403':
        content:
          application/json:
            schema:

```

```

        $ref: '#/components/schemas/Error'
      description: Forbidden
    '404':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Not found
  /api/reserve/{id}/medicine/{med_pk}/usages:
    get:
      operationId: api_reserve_medicine_usages_list
      summary: Get usage of medicine in reserve
      parameters:
        - in: path
          name: id
          schema:
            type: integer
          required: true
        - in: path
          name: med_pk
          schema:
            type: integer
          required: true
      tags:
        - api
      responses:
        '200':
          content:
            application/json:
              schema:
                type: array
                items:
                  $ref: '#/components/schemas/ReserveActivity'
          description: ''
        '401':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
          description: Not authenticated
        '403':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
          description: Forbidden
        '404':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
          description: Not found
  /api/reserve/{id}/medicine/update/{med_pk}:
    put:

```

```

operationId: api_reserve_medicine_update_update
summary: Update a medicine in a reserve
parameters:
- in: path
  name: id
  schema:
    type: integer
    required: true
- in: path
  name: med_pk
  schema:
    type: integer
    required: true
tags:
- api
requestBody:
  content:
    application/json:
      schema:
        $ref: '#/components/schemas/ReserveMedicine'
    application/x-www-form-urlencoded:
      schema:
        $ref: '#/components/schemas/ReserveMedicine'
    multipart/form-data:
      schema:
        $ref: '#/components/schemas/ReserveMedicine'
  required: true
responses:
  '200':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/ReserveMedicine'
    description: ''
  '400':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Bad request
  '401':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Not authenticated
  '403':
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
    description: Forbidden
  '404':
    content:

```

```

        application/json:
          schema:
            $ref: '#/components/schemas/Error'
        description: Not found
/api/reserve/add_medicine:
  post:
    operationId: api_reserve_add_medicine_create
    summary: Add a medicine to a reserve
    tags:
      - api
    requestBody:
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/ReserveMedicine'
        application/x-www-form-urlencoded:
          schema:
            $ref: '#/components/schemas/ReserveMedicine'
        multipart/form-data:
          schema:
            $ref: '#/components/schemas/ReserveMedicine'
      required: true
    responses:
      '201':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/ReserveMedicine'
          description: ''
      '400':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
          description: Bad request
      '401':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
          description: Not authenticated
/api/reserve/create:
  post:
    operationId: api_reserve_create_create
    summary: Create a new reserve
    tags:
      - api
    requestBody:
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Reserve'
        application/x-www-form-urlencoded:
          schema:

```

```

        $ref: '#/components/schemas/Reserve'
    multipart/form-data:
        schema:
            $ref: '#/components/schemas/Reserve'
    required: true
responses:
    '201':
        content:
            application/json:
                schema:
                    $ref: '#/components/schemas/Reserve'
        description: ''
    '400':
        content:
            application/json:
                schema:
                    $ref: '#/components/schemas/Error'
        description: Bad request
    '401':
        content:
            application/json:
                schema:
                    $ref: '#/components/schemas/Error'
        description: Not authenticated
/api/reserve/delete/{id}:
    delete:
        operationId: api_reserve_delete_destroy
        summary: Delete a reserve
        parameters:
            - in: path
              name: id
              schema:
                  type: integer
              required: true
        tags:
            - api
        responses:
            '204':
                description: Deleted successfully
            '400':
                content:
                    application/json:
                        schema:
                            $ref: '#/components/schemas/Error'
                description: Bad request
            '401':
                content:
                    application/json:
                        schema:
                            $ref: '#/components/schemas/Error'
                description: Not authenticated
            '403':
                content:
                    application/json:

```

```

        schema:
          $ref: '#/components/schemas/Error'
      description: Forbidden
    '404':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Not found
  /api/reserve/update/{id}:
    put:
      operationId: api_reserve_update_update
      summary: Update a reserve
      parameters:
        - in: path
          name: id
          schema:
            type: integer
          required: true
      tags:
        - api
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Reserve'
          application/x-www-form-urlencoded:
            schema:
              $ref: '#/components/schemas/Reserve'
          multipart/form-data:
            schema:
              $ref: '#/components/schemas/Reserve'
        required: true
      responses:
        '200':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Reserve'
          description: ''
        '400':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
          description: Bad request
        '401':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
          description: Not authenticated
        '403':
          content:

```

```

        application/json:
          schema:
            $ref: '#/components/schemas/Error'
          description: Forbidden
      '404':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
            description: Not found
/api/reserves:
  get:
    operationId: api_reserves_list
    summary: List all reserves of the authenticated user
    tags:
      - api
    responses:
      '200':
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: '#/components/schemas/Reserve'
            description: ''
      '401':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
            description: Not authenticated
/api/schema/:
  get:
    operationId: api_schema_retrieve
    description: |-
      OpenApi3 schema for this API. Format can be selected via content negotiation.

      - YAML: application/vnd.oai.openapi
      - JSON: application/vnd.oai.openapi+json
    parameters:
      - in: query
        name: format
        schema:
          type: string
          enum:
            - json
            - yaml
      - in: query
        name: lang
        schema:
          type: string
          enum:
            - af
            - ar

```

- ar-dz
- ast
- az
- be
- bg
- bn
- br
- bs
- ca
- ckb
- cs
- cy
- da
- de
- dsb
- el
- en
- en-au
- en-gb
- eo
- es
- es-ar
- es-co
- es-mx
- es-ni
- es-ve
- et
- eu
- fa
- fi
- fr
- fy
- ga
- gd
- gl
- he
- hi
- hr
- hsb
- hu
- hy
- ia
- id
- ig
- io
- is
- it
- ja
- ka
- kab
- kk
- km
- kn
- ko

- ky
- lb
- lt
- lv
- mk
- ml
- mn
- mr
- ms
- my
- nb
- ne
- nl
- nn
- os
- pa
- pl
- pt
- pt-br
- ro
- ru
- sk
- sl
- sq
- sr
- sr-latn
- sv
- sw
- ta
- te
- tg
- th
- tk
- tr
- tt
- udm
- ug
- uk
- ur
- uz
- vi
- zh-hans
- zh-hant

tags:

- api

security:

- {}

responses:

 '200':

 content:

 application/vnd.oai.openapi:

 schema:

 type: object

 additionalProperties: {}

```

    application/yaml:
      schema:
        type: object
        additionalProperties: {}
    application/vnd.oai.openapi+json:
      schema:
        type: object
        additionalProperties: {}
    application/json:
      schema:
        type: object
        additionalProperties: {}
  description: ''
/api/usage:
  get:
    operationId: api_usage_list
    summary: List all usage records
    tags:
      - api
    responses:
      '200':
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: '#/components/schemas/Usage'
        description: ''
/api/usage/{id}:
  get:
    operationId: api_usage_retrieve
    summary: Retrieve a specific usage record
    parameters:
      - in: path
        name: id
        schema:
          type: integer
        required: true
    tags:
      - api
    responses:
      '200':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Usage'
        description: ''
      '404':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
        description: Not found
/api/usage/{id}/dosages:

```

```

get:
  operationId: api_usage_dosages_list
  summary: List all dosages for a specific usage
  parameters:
    - in: path
      name: id
      schema:
        type: integer
      required: true
  tags:
    - api
  responses:
    '200':
      content:
        application/json:
          schema:
            type: array
            items:
              $ref: '#/components/schemas/Dosage'
      description: ''
    '404':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Not found
/api/usage/create:
  post:
    operationId: api_usage_create_create
    summary: Create a new usage record
    tags:
      - api
    requestBody:
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Usage'
        application/x-www-form-urlencoded:
          schema:
            $ref: '#/components/schemas/Usage'
        multipart/form-data:
          schema:
            $ref: '#/components/schemas/Usage'
    responses:
      '201':
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Usage'
        description: ''
      '400':
        content:
          application/json:
            schema:

```

```

        $ref: '#/components/schemas/Error'
      description: Bad request
    '401':
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
      description: Not authenticated
  /api/usage/delete/{id}:
    delete:
      operationId: api_usage_delete_destroy
      summary: Delete a usage record
      parameters:
        - in: path
          name: id
          schema:
            type: integer
          required: true
      tags:
        - api
      responses:
        '204':
          description: Deleted successfully
        '401':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
          description: Not authenticated
        '404':
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
          description: Not found
  /api/usage/update/{id}:
    put:
      operationId: api_usage_update_update
      summary: Update an existing usage record
      parameters:
        - in: path
          name: id
          schema:
            type: integer
          required: true
      tags:
        - api
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Usage'
          application/x-www-form-urlencoded:
            schema:

```

```

        $ref: '#/components/schemas/Usage'
    multipart/form-data:
        schema:
            $ref: '#/components/schemas/Usage'
responses:
    '200':
        content:
            application/json:
                schema:
                    $ref: '#/components/schemas/Usage'
        description: ''
    '400':
        content:
            application/json:
                schema:
                    $ref: '#/components/schemas/Error'
        description: Bad request
    '401':
        content:
            application/json:
                schema:
                    $ref: '#/components/schemas/Error'
        description: Not authenticated
    '404':
        content:
            application/json:
                schema:
                    $ref: '#/components/schemas/Error'
        description: Not found
components:
    schemas:
        AgeUnitEnum:
            enum:
                - y
                - m
                - d
            type: string
            description: |-
                * `y` - years
                * `m` - months
                * `d` - days
        BlankEnum:
            enum:
                - ''
    Categories:
        type: object
        properties:
            id:
                type: integer
                readOnly: true
            name:
                type: string
                maxLength: 100
        required:

```

```

- id
- name
Category:
  type: object
  properties:
    id:
      type: integer
      readOnly: true
    name:
      type: string
      maxLength: 100
    medicines:
      type: array
      items:
        type: string
        readOnly: true
  required:
    - id
    - medicines
    - name
Dosage:
  type: object
  properties:
    id:
      type: integer
      readOnly: true
    amount:
      type: number
      format: double
    time:
      $ref: '#/components/schemas/TimeEnum'
    usage:
      type: integer
  required:
    - amount
    - id
    - time
    - usage
Error:
  type: object
  properties:
    detail:
      type: string
      description: Error message
  required:
    - detail
Medicine:
  type: object
  properties:
    id:
      type: integer
      readOnly: true
    name:
      type: string

```

```

        maxLength: 100
    amount:
        type: number
        format: double
    unit:
        $ref: '#/components/schemas/UnitEnum'
    is_global:
        type: boolean
    added_by:
        type: integer
    categories:
        type: array
        items:
            type: integer
    required:
        - added_by
        - amount
        - categories
        - id
        - name
        - unit
MedicineIntake:
    type: object
    properties:
        reserve_medicine:
            type: integer
        usage:
            type: integer
        duration_from:
            type: string
            format: date
        duration_to:
            type: string
            format: date
            nullable: true
        morning_time:
            type: string
            format: time
            nullable: true
        afternoon_time:
            type: string
            format: time
            nullable: true
        evening_time:
            type: string
            format: time
            nullable: true
        night_time:
            type: string
            format: time
            nullable: true
        last_intake:
            type: string
            format: date-time

```

```

    required:
      - reserve_medicine
      - usage
  NullEnum:
    enum:
      - null
  PendingMedicine:
    type: object
    properties:
      id:
        type: integer
        readOnly: true
      is_closed:
        type: boolean
      status:
        $ref: '#/components/schemas/StatusEnum'
      last_modified:
        type: string
        format: date-time
        readOnly: true
      comments:
        type: string
      review_comments:
        type: string
      medicine:
        type: integer
    required:
      - id
      - last_modified
      - medicine
  Profile:
    type: object
    properties:
      id:
        type: integer
        readOnly: true
      username:
        type: string
        readOnly: true
      age:
        type: integer
        maximum: 2147483647
        minimum: -2147483648
        nullable: true
      morning_time:
        type: string
        format: time
        nullable: true
      afternoon_time:
        type: string
        format: time
        nullable: true
      evening_time:
        type: string

```



```

        format: time
        nullable: true
    night_time:
        type: string
        format: time
        nullable: true
    is_administrator:
        type: boolean
    role:
        $ref: '#/components/schemas/RoleEnum'
required:
- id
- username
Reserve:
    type: object
    properties:
        id:
            type: integer
            readOnly: true
        name:
            type: string
            maxLength: 100
    required:
    - id
    - name
ReserveActivity:
    type: object
    properties:
        id:
            type: integer
            readOnly: true
        consumed_amount:
            type: number
            format: double
        date_time:
            type: string
            format: date-time
            readOnly: true
        reserve_medicine:
            type: integer
    required:
    - consumed_amount
    - date_time
    - id
    - reserve_medicine
ReserveMedicine:
    type: object
    properties:
        id:
            type: integer
            readOnly: true
        amount:
            type: number
            format: double

```

```

    medicine:
      type: integer
    reserve:
      type: integer
  required:
    - amount
    - id
    - medicine
    - reserve
RoleEnum:
  enum:
    - guest
    - admin
    - client
  type: string
  description: |-
    * `guest` - guest
    * `admin` - admin
    * `client` - client
StatusEnum:
  enum:
    - a
    - r
    - p
  type: string
  description: |-
    * `a` - approved
    * `r` - rejected
    * `p` - pending
TimeEnum:
  enum:
    - Morning
    - Afternoon
    - Evening
    - Night
  type: string
  description: |-
    * `Morning` - 06:00-12:00
    * `Afternoon` - 12:00-17:00
    * `Evening` - 17:00-21:00
    * `Night` - 21:00-00:00
UnitEnum:
  enum:
    - ml
    - g
    - pcs
  type: string
  description: |-
    * `ml` - milliliters
    * `g` - grams
    * `pcs` - pieces
Usage:
  type: object
  properties:

```

```

    id:
      type: integer
      readOnly: true
    age_from:
      type: integer
      maximum: 2147483647
      minimum: -2147483648
      nullable: true
    age_to:
      type: integer
      maximum: 2147483647
      minimum: -2147483648
      nullable: true
    age_unit:
      nullable: true
      oneOf:
        - $ref: '#/components/schemas/AgeUnitEnum'
        - $ref: '#/components/schemas/BlankEnum'
        - $ref: '#/components/schemas/NullEnum'
    comments:
      type: string
    duration_days:
      type: integer
      maximum: 2147483647
      minimum: -2147483648
      nullable: true
    hours_between:
      type: integer
      maximum: 2147483647
      minimum: -2147483648
      nullable: true
    medicine:
      type: integer
      nullable: true
  required:
    - id
User:
  type: object
  properties:
    id:
      type: integer
      readOnly: true
    username:
      type: string
      description: Required. 150 characters or fewer. Letters, digits and @/./+/-/_ only.
      pattern: ^[\w.@+-]+$
      maxLength: 150
    email:
      type: string
      format: email
    password:
      type: string
      maxLength: 128

```

required:
- id
- password
- username

5. Išvados

Šiam projektui atlikti buvo naudojama Django serverinė dalis, React klientinė dalis bei PostgreSQL duomenų bazė. Projekto darymo metu buvo susipažinta su bei išmokta panaudoti OpenAPI specifikaciją ir JWT tokenus.

Klientinės dalies įgyvendinimui pritrūko laiko, todėl įgyvendintas tik prisijungimas ir vienas API. Tačiau klientinės dalies įgyvendinime galima pamatyti panaudotus modalus, ikonas, prisijungimą ir žetono atnaujinimą su JWT tokenais bei skirtingą elementų kiekį navigacijos juostoje, priklausomai nuo rolės.